Formula 1 Podium Prediction

CSEN 903 Advanced Computer Lab, Winter 2026: Milestone 1 Analytical Report

Team 121

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Kaggle notebook: https://www.kaggle.com/code/omaryasser102/milestone-1-121-acl

GitHub repository: https://github.com/Omarryasserr102/ACL2-Team121-F1-M1

Item	Choice
Task	Binary classification: does a driver finish on the podium (`positionOrder` ≤ 3)?
Unit of analysis	One driver entering one race; (`raceId, driverId`) is unique
Prediction cutoff	After qualifying and the final grid are known, before the race starts
Data	Kaggle / Ergast F1 World Championship 1950–2024, 14 relational tables
Modelling table	24,878 race starts, 20 pre-race features, 13.6% podiums
Split (by season)	Train 1950–2019 (22,804 rows) · Validation 2020–2021 (759) · Test 2022–2024 (1,315)
Metrics	Primary PR-AUC, secondary ROC-AUC, F1 at a validation-tuned threshold
Selected model	Logistic regression (baseline): test PR-AUC 0.700, ROC-AUC 0.930, F1 0.660

This report summarises the analysis in our run-all notebook. Every number and figure below is taken from that notebook's output. The notebook also holds the full code, intermediate tables and additional plots.

1. Data audit, cleaning and exploratory analysis

1.1 Before cleaning

The raw tables were loaded unchanged, keeping the literal `\N` placeholders visible. `pandas` does not recognise them as missing, so we counted them ourselves: 160,076 sentinels in six tables, 122,887 of them in `results` (times, fastest laps, finishing position for non-classified drivers). The `q2`/`q3` qualifying times are missing by design, because only drivers who advance set them. The most important finding for modelling is that **qualifying data exists for only 43.9% of races**, none before 1994 and only patchy coverage until 2002 (Figure 1). This is why the starting grid, which exists for every season, is our main starting-position feature and qualifying is treated as optional, with missing-value indicators. About 12.7% of result rows are podiums, so the positive class is the minority.

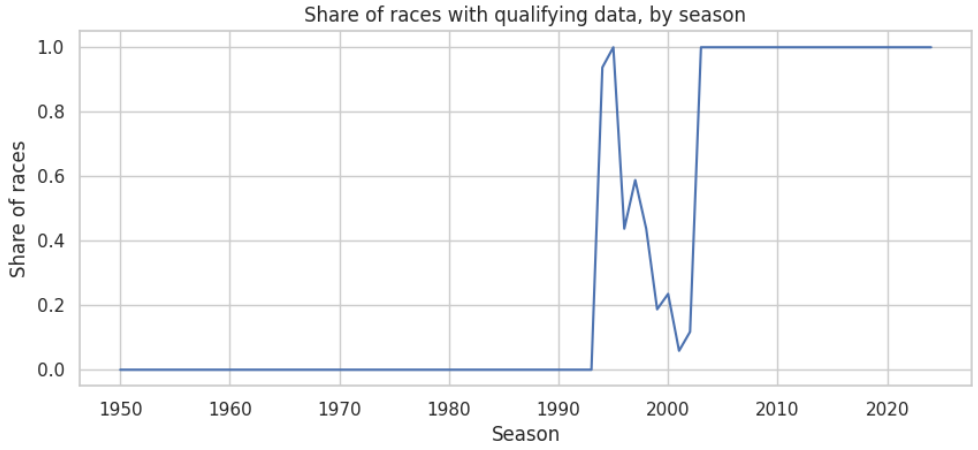


Figure 1. *Share of races per season with a qualifying record: none before 1994, patchy between 1996 and 2002, complete from 2003.*

1.2 Audit and cleaning

- **Keys:** all 14 primary keys are unique and non-null, and all 22 foreign-key relationships reconcile with their parent tables (0 unmatched rows).
- **Sentinels:** `\N` was replaced with NaN on copies; the raw CSVs stay immutable. Figure 2 confirms that every sentinel became exactly one NaN, so nothing was lost or invented.
- **Semantic typing:** race dates and birth dates became datetimes, and lap, qualifying and pit-stop durations became numeric seconds (e.g. `1:27.452` → 87.452).
- **Grain:** 176 result rows (85 driver-race pairs) are 1950s shared drives. Rule: keep the car the driver *started* in (grid > 0), then the better finish, then the lower `resultId`. Our features describe the starting situation, which belongs to that car. 91 rows were removed, leaving one row per `(raceId, driverId)`.
- **Modelling population:** drivers who took the start from a grid slot (grid > 0, not failed to qualify, withdrawn or excluded), giving 24,878 rows with 13.6% podiums.

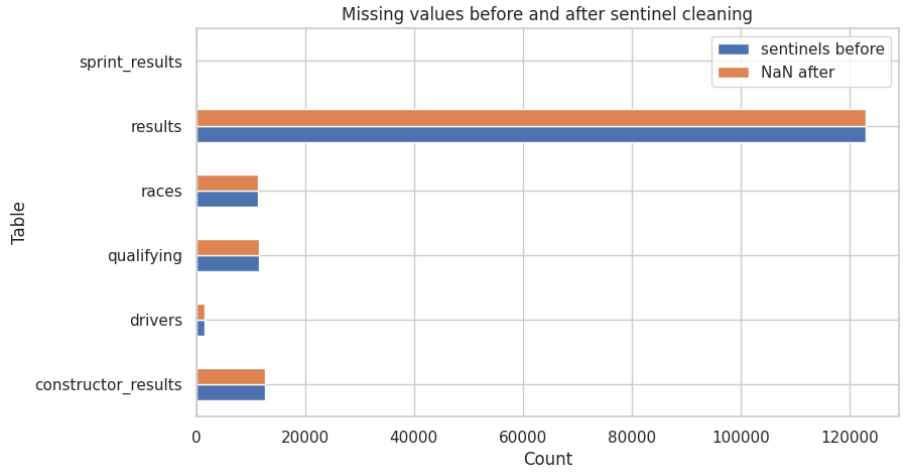


Figure 2. *Sentinel counts before cleaning vs NaN counts after cleaning, per table. The bars match exactly.*

1.3 Leakage control

Only information known before the start may be a feature. We use `grid` and qualifying (both fixed before the race), static driver and circuit attributes, and form features computed from *earlier races only*. The final position, points, laps, times, fastest laps, status, lap times, pit stops and the current race's standings are excluded. Sprint results are excluded because they exist only from 2021 and their place relative to qualifying changed between seasons.

2. Data-engineering questions

All three questions use multi-hop joins on the cleaned, one-row-per-driver-race table. Because they describe completed races, they may use final results and status; none of these fields is a model feature.

Q1. Which circuits convert a front-row start into a podium most often, before and after 2014?

Join path: results → races → circuits, plus qualifying. To compare like with like, we only rank circuits with at least 10 front-row starts in *both* eras (16 circuits, 2,243 front-row grid starts in total).

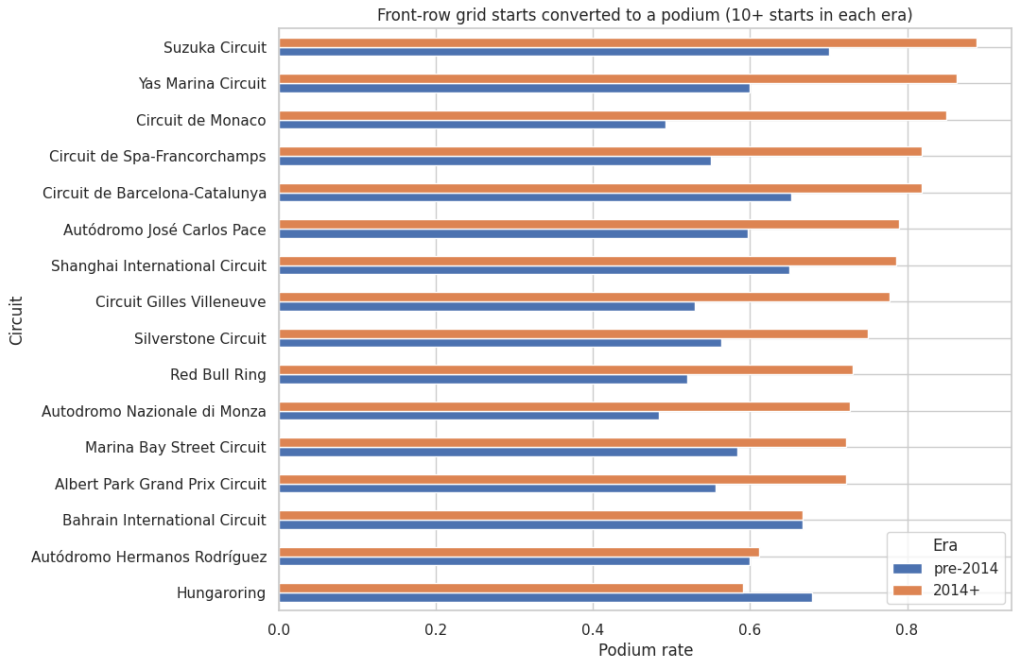


Figure 3. *Podium rate of front-row (grid P1–P2) starters per circuit, before 2014 and from 2014 onwards.*

Answer. Suzuka converts a front-row start into a podium most often in both eras (70% before 2014, 89% after). Below the top, the ranking changes markedly in the hybrid era. **Monaco** climbs from 15th (49%) to 3rd (85%), and Spa, Monza and Circuit Gilles Villeneuve gain 24–27 percentage points. The **Hungaroring** falls from 2nd (68%) to last (59%) and is the only circuit whose rate dropped. Overall, front-row conversion rises from 65% to 76%, consistent with the long spells of single-team dominance after 2014.

Grid vs qualifying. The two differ for 2,509 of the 10,494 rows where both exist (penalties, pit-lane starts). Measured by qualifying position, front-row conversion also rises after 2014 (64.7% → 73.7%). Because qualifying records only become complete in 2003, the per-circuit ranking uses the grid.

Q2. Do drivers podium more often at home, after controlling for starting position?

Join path: results → races → circuits (country) and results → drivers (nationality). Nationalities were mapped to circuit countries with an explicit dictionary, and unmappable ones (e.g. Chilean, Rhodesian) were reported and excluded. To control for starting position, home and away drivers are compared *within the same grid band*, and the band differences are averaged with band-size weights.

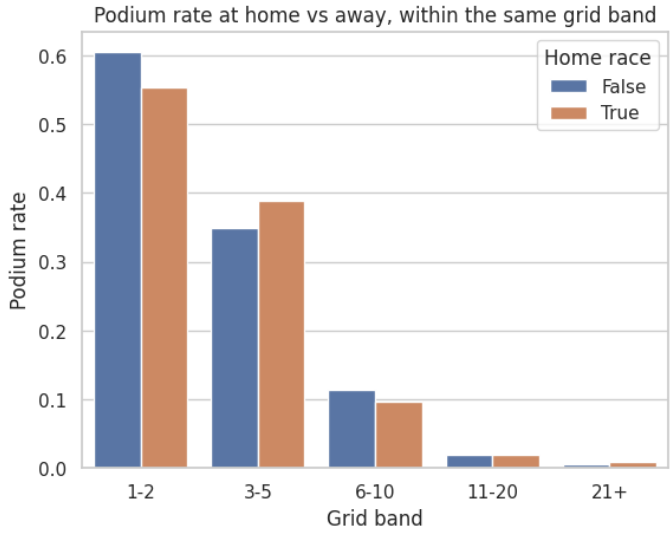


Figure 4. *Podium rate at home vs away within each starting-grid band.*

Answer. No. The grid-adjusted home-minus-away podium rate is –**0.28 percentage points**, essentially zero. The sign even flips between bands: home drivers do slightly worse from the front row (55.3% vs 60.5%) and slightly better from P3–P5 (38.9% vs 34.9%), which is noise rather than a consistent effect. The home samples in the front bands are small (179 home front-row starts), and grid bands do not remove every driver or team difference.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Join path: results → status → constructors, plus races. A mechanical retirement is a race start that ended because a car component failed. The `status` text was classified with an explicit keyword list (engine, gearbox, hydraulics, power unit, overheating, etc.). Crashes, driver reasons, regulation decisions, pit-equipment issues (e.g. fuel rig) and ambiguous reasons (e.g. "Retired", "Vibrations", wing damage) are excluded; the notebook lists every excluded status. The denominator is race starts, not entries.

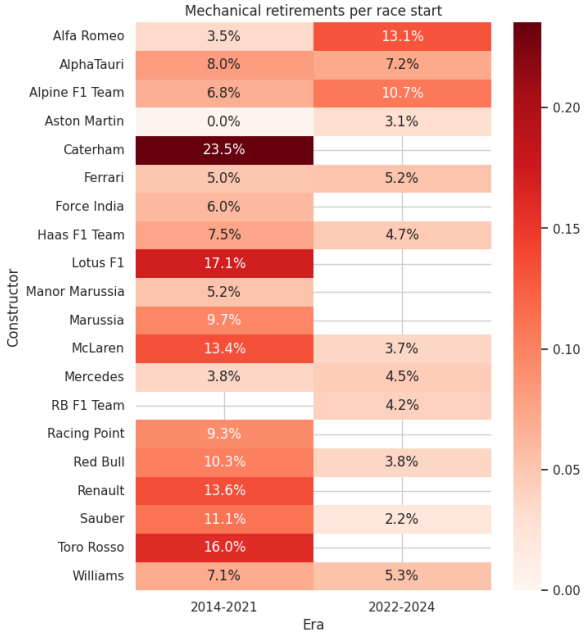


Figure 5. *Mechanical retirements per race start by constructor and era.*

Answer. In **2014–2021** the highest rates belong to **Caterham** (23.5%, but from only 34 starts), **Lotus F1** (17.1%), **Toro Rosso** (16.0%), **Renault** (13.6%) and **McLaren** (13.4% from 320 starts, the McLaren-Honda years). In **2022–2024** the leaders are **Alfa Romeo** (13.1%) and **Alpine** (10.7%), clearly ahead of the rest. Reliability improved markedly in the ground-effect era, with most teams at or below about 5%. Small denominators make some rates unstable, and rebranded teams appear as separate constructors.

3. Selected features and their validity at the pre-race cutoff

Group	Features	Why it is known before the race
Grid / qualifying	grid_position, quali_position, quali_time_gap_to_pole	Final starting slot and qualifying result are fixed before lights out (qualifying missing before 1994 and patchy until 2002)
Driver form	career_races, career_wins, career_podiums, avg_finish_last_5, podium_rate_last_10, wins_this_season_so_far	Computed from earlier races only: cumulative counts exclude the current race and rolling windows are shifted by one race
Constructor form	constructor_wins_total, constructor_podium_rate, constructor_wins_this_season	Team results from earlier races only, built at one row per team per race
Driver profile	driver_age_at_race, is_home_race	Date of birth, nationality and venue are known in advance
Race context	year, round, starter_count	Calendar facts; the number of starters is known once the grid is set
Circuit	lat, alt, country	Static circuit attributes (track length and type do not exist in the data, so they were not invented)

Table 1. *The 20 model features and why each is valid at the cutoff.*

Evidence before training (training seasons only, so validation and test cannot influence the choice). The strongest single signals are qualifying position (correlation with podium –0.48), recent podium rate (+0.47), grid position (–0.44), constructor wins this season (+0.37), driver season wins (+0.36) and recent average finish (–0.36). Figure 7 shows the podium rate rising steadily across the ranges of the form features. Driver profile, race context and circuit features have near-zero correlations. We kept them as context, because era and field size change how hard a podium is, and tested them in the ablation (Section 6).

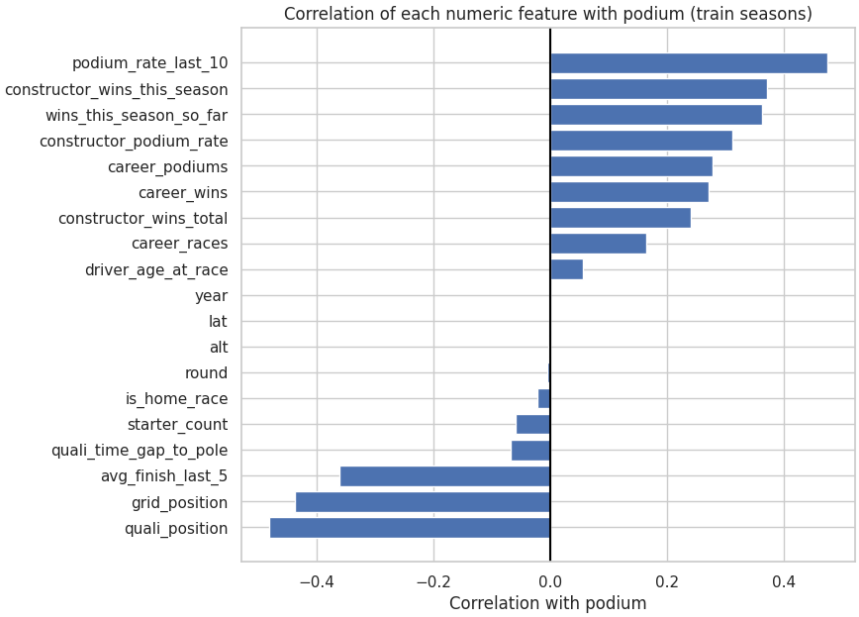


Figure 6. *Correlation of each numeric feature with podium (train seasons).*

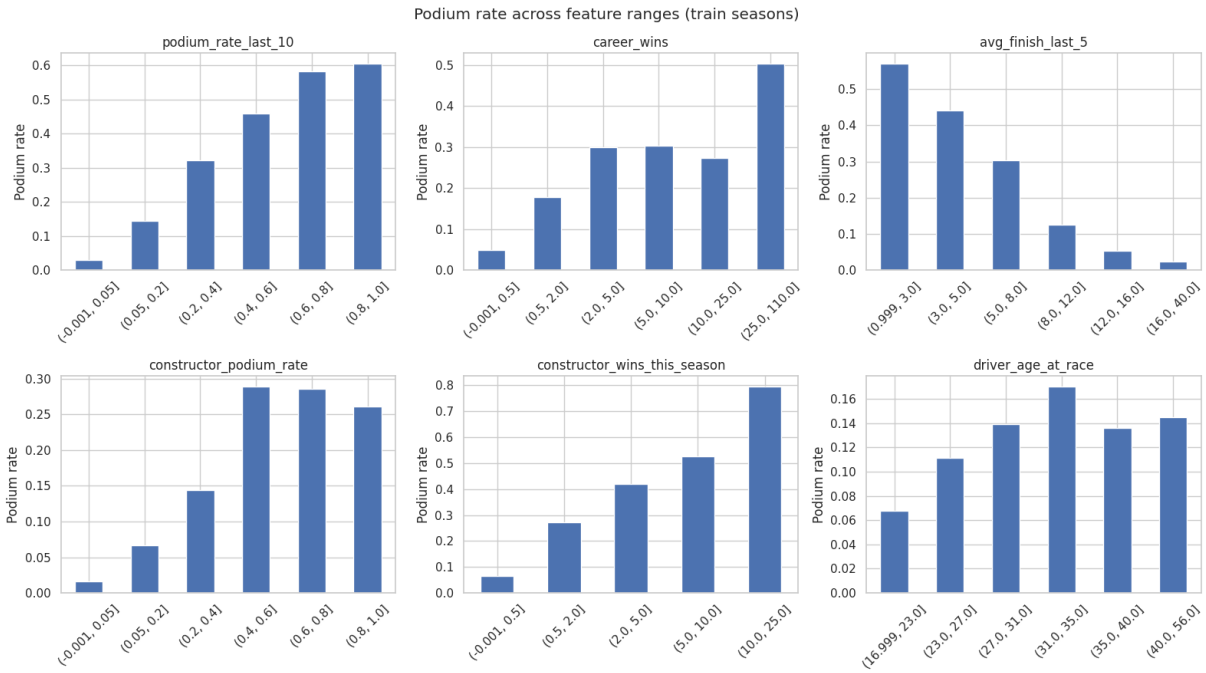


Figure 7. *Podium rate across the ranges of six form and profile features (train seasons).*

4. Preprocessing

Three steps are needed: imputation (qualifying features are missing for 58% of rows), one-hot encoding of `country`, and standardisation. We compared three orders with the same logistic regression, fitted on the training seasons only:

Pipeline	Order	Validation PR-AUC	Validation ROC-AUC
A (chosen)	impute (median) → encode → scale, per column type	0.721	0.929
B	encode → impute (median) → scale, one imputer for all columns	0.718	0.929
C	scale → impute (mean, i.e. 0 in scaled units)	0.718	0.928

Table 2. *Preprocessing-order experiment.*

The spread between the three is only 0.004, far smaller than the uncertainty of PR-AUC on 759 validation rows (bootstrap intervals are about 0.17 wide). The step order is therefore practically irrelevant here, because all three handle missing qualifying values the same way: a fill value plus a missing-indicator column. We kept Pipeline A, which has the best score and the most standard design.

Effect on the data. Median imputation fills 14,532 of the 22,804 training values of `quali_position` with 11. The mean barely moves (11.3 → 11.1), but the standard deviation drops from 6.4 to 3.8 and a large spike appears (Figure 8). The indicator column lets the model separate real from filled values. Scaling changes only the units: every column ends with mean 0 and standard deviation 1, and the shapes are unchanged, so large-unit features such as `career_wins` (0–91) no longer dominate small ones.

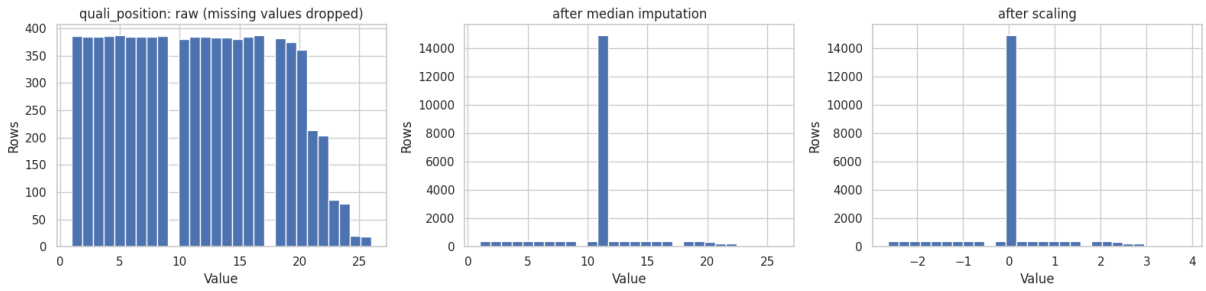


Figure 8. *quali_position in the training seasons: observed values, after median imputation, and after scaling.*

5. Predictive modelling

5.1 Models, metrics and tuning

All models use the same rows, features and preprocessing. **Logistic regression** is the baseline: a single weighted sum of the features passed through a sigmoid. The **random forest** averages many decision trees trained on bootstrapped rows and random feature subsets, so it can capture interactions. The **shallow neural network** (Keras) has one or two ReLU hidden layers and a sigmoid output, trained with binary cross-entropy, Adam and early stopping on the latest 15% of the training seasons.

Metric choice. Podiums are only about 14% of rows, so the **primary metric is PR-AUC**. It measures how well real podium finishers are ranked above everyone else, and unlike ROC-AUC it is not inflated by the many easy negatives. **ROC-AUC** is the secondary metric for overall separation. **F1** is reported at the threshold that maximises F1 on the validation set. Thresholds and hyperparameters were chosen on validation only; the test set was used once, at the end.

Model	Candidate settings (validation PR-AUC)	Chosen
Logistic regression	C = 0.1 (0.725) · C = 1 (0.721) · C = 10 (0.720)	C = 0.1
Random forest	depth 6 / leaf 4 (0.697) · depth 12 / leaf 4 (0.726) · unlimited depth / leaf 10 (0.750)	unlimited depth, leaf 10
Neural network	[32] (0.723) · [64, 32] (0.734) · [32] balanced (0.704) · [64, 32] balanced (0.663)	[64, 32], no class weight

Table 3. *Hyperparameter and architecture comparison (validation only).*

Logistic regression is insensitive to regularisation. The forest underfits with shallow trees and is best with fully grown trees that need at least 10 rows per leaf. For the network, two layers beat one, and class weighting hurt ranking quality, so the unweighted network was kept.

5.2 Results

Model	Split	ROC-AUC	PR-AUC	F1	Precision	Recall	Threshold
Logistic regression	train	0.900	0.561	0.579	0.471	0.750	0.65
	validation	0.928	0.725	0.719	0.662	0.786	
	test	0.930	0.700	0.660	0.537	0.858	
Random forest	train	0.953	0.755	0.677	0.661	0.694	0.70
	validation	0.930	0.750	0.722	0.735	0.709	
	test	0.933	0.695	0.665	0.637	0.696	
Neural network	train	0.903	0.578	0.584	0.476	0.756	0.23
	validation	0.925	0.733	0.714	0.667	0.769	
	test	0.920	0.659	0.631	0.583	0.686	

Table 4. *Model comparison on train, validation and the unseen test set (accuracy is in the notebook).*

Model selection. The random forest has the highest validation PR-AUC (0.750), but the validation set holds only about 117 podiums. A paired bootstrap (1,000 resamples) shows the forest's advantage over the baseline has a 95% interval of −0.006 to +0.059, which includes zero. Our rule picks the simplest model that is not significantly worse than the best one, so the **logistic-regression baseline is selected**: no more complex model clearly earned its extra complexity. On the test set it is in fact the best model (PR-AUC 0.700 vs 0.695 and 0.659).

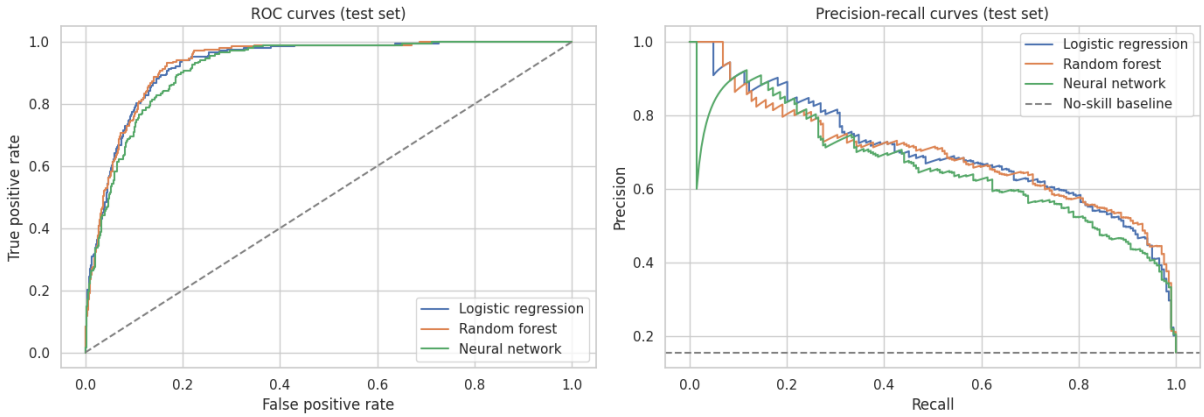


Figure 9. *ROC and precision-recall curves of the three tuned models on the test set (2022–2024).*

- **Train below validation** for logistic regression and the network (PR-AUC about 0.57 vs 0.73): the training seasons include the 1950s–1980s, with irregular fields, frequent retirements and no qualifying data, where podiums are far harder to predict before the race.
- **Validation to test:** PR-AUC drops by 0.03–0.07 while ROC-AUC stays at 0.92–0.93. The 2022 regulation change makes the test seasons a new car era. The network generalises worst (0.733 → 0.659).
- **Thresholds differ from 0.5:** 0.65 (logistic regression) and 0.70 (forest), because balanced class weights deliberately push their probabilities up, and 0.23 for the unweighted network. The F1-optimal threshold is generally not 0.5, so it is tuned on validation.
- **Random forest train vs validation** (0.755 vs 0.750): because the training seasons are harder to predict, train ≈ validation means the forest fits the training data much more tightly than the other models; the leaf-size limit keeps this from hurting validation.

5.3 Model limitations

- **Logistic regression** is linear in the features and cannot learn interactions (e.g. a strong team starting from the front row) unless they are engineered. Its coefficients are also distorted by correlated inputs (Section 7).
- **Random forest** can memorise training seasons, which is controlled here by the leaf size, and cannot extrapolate beyond the feature ranges seen in training.
- **Neural network** is the least interpretable, needs scaling and early stopping, depends on its random initialisation (we fix seeds and use deterministic operations). Its early-stopping loss was lowest at the 2nd epoch (training stopped at epoch 17 and restored those weights), so it gains little from its extra capacity on this data.

6. Feature-group ablation

Each feature group was removed in turn and the selected model retrained with the same pipeline (validation 2020–2021).

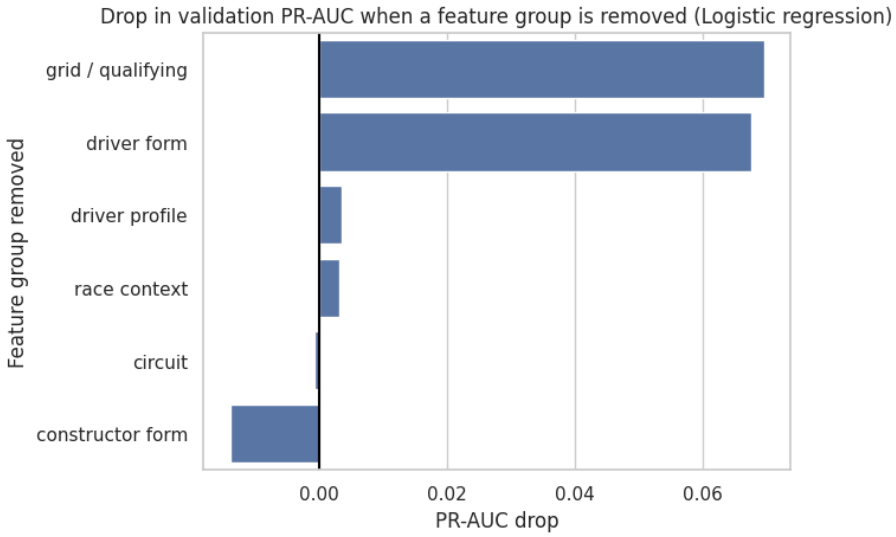


Figure 10. *Drop in validation PR-AUC when each feature group is removed (logistic regression).*

- Removing **grid / qualifying** lowers PR-AUC by 0.070 (and ROC-AUC by 0.040), and removing **driver form** lowers PR-AUC by 0.068: both groups are indispensable. Grid carries most of the overall separation, while form mainly helps rank the top candidates.
- **Driver profile, race context and circuit** change PR-AUC by less than 0.004, which is noise on 759 rows.
- **Constructor form** slightly *improves* PR-AUC when removed (+0.014). Team strength is already reflected in the grid slot and the driver's recent results, and two of the three constructor features are all-time totals, which are stale for historic teams that are weak today. The effect is well inside the validation uncertainty, so the group was kept; a rolling constructor form is a natural improvement.

This matches the pre-training evidence for the strong groups. The weak context groups contribute little, as their near-zero correlations suggested.

7. Model explainability

Important: these explanations are not causal. They describe what the fitted model *relies on* to make its predictions. A feature being important does not mean it causes podium finishes, only that the model uses it.

7.1 Global explanations

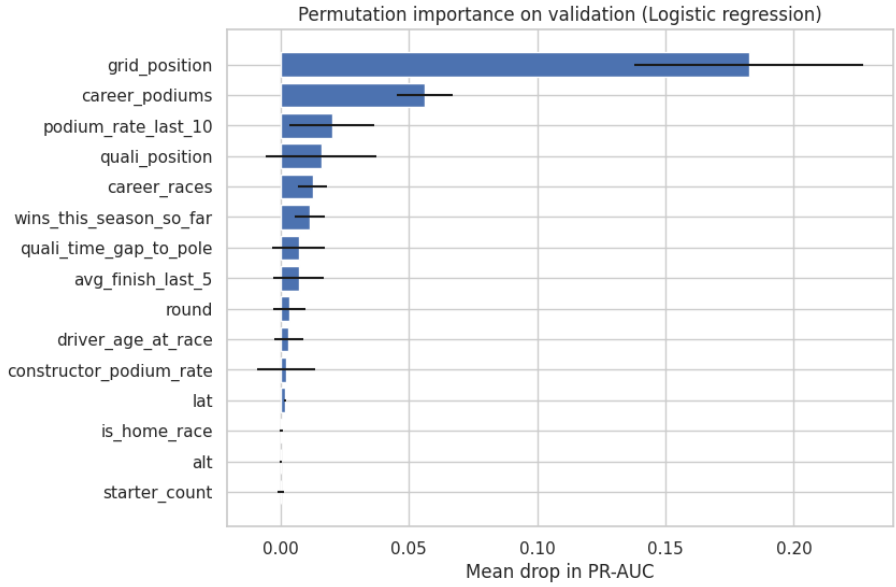


Figure 11. *Permutation importance on the validation set (mean drop in PR-AUC over 10 shuffles, with standard deviation).*

Permutation importance. `grid_position` dominates (0.18), more than three times any other feature, followed by `career_podiums` (0.056) and recent podium rate (0.020). `quali_position` scores lower than its correlation suggests because it overlaps with the grid: when one is shuffled, the other still carries the information. Features with importance near zero or slightly negative (`career_wins`, constructor wins) are related to podiums (Section 3), but the model gets the same information from correlated features.

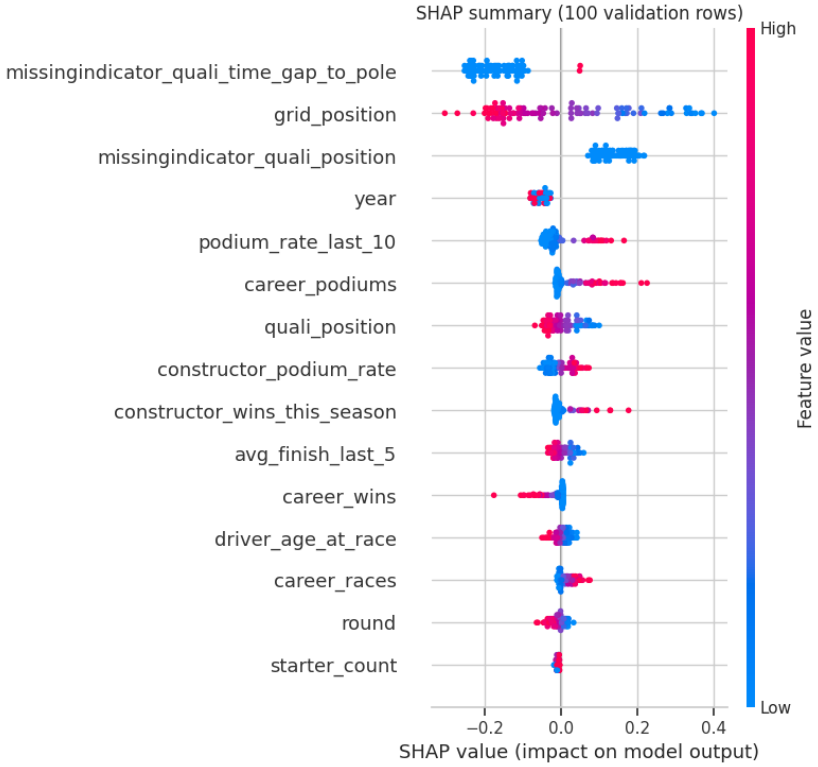


Figure 12. *SHAP summary for 100 validation rows (feature values in scaled units).*

SHAP. A front-of-grid start pushes the probability up strongly, and strong recent and career form raise it further, so the two methods agree on the main story. One pattern needs care. Every 2020–2021 validation race has qualifying data, so the two missing-indicator columns take the same value for every row explained here. Their SHAP values say nothing about the individual driver: they measure "qualifying recorded" against a training background where about 58% of rows have no qualifying data, i.e. an **era offset**. This is also why they are near zero in permutation importance, since shuffling a constant column changes nothing. The two columns are almost identical in training (missing for 58.4% vs 59.0% of rows) but receive opposite-sign weights, so they must be read together; their net effect is about –0.08 for the Sainz record below. The negative push from `career_wins` is a correction against the correlated `career_podiums`. The SHAP base value (0.37) is above the 14% podium share because the balanced class weights push the predicted probabilities up.

7.2 Local explanations

Correct prediction: Lewis Hamilton, 2020 Bahrain GP (started 1st, podium, predicted 0.991). From an average prediction of 0.37, SHAP shows pole position adding the most (+0.24), followed by his career podium count (+0.16) and recent podium rate (+0.11), with the qualifying-indicator era offset (+0.13) in between; the `career_wins` correction (–0.11) pulls the other way. LIME agrees: its largest bar is the front-row grid slot, followed by recent podium rate.

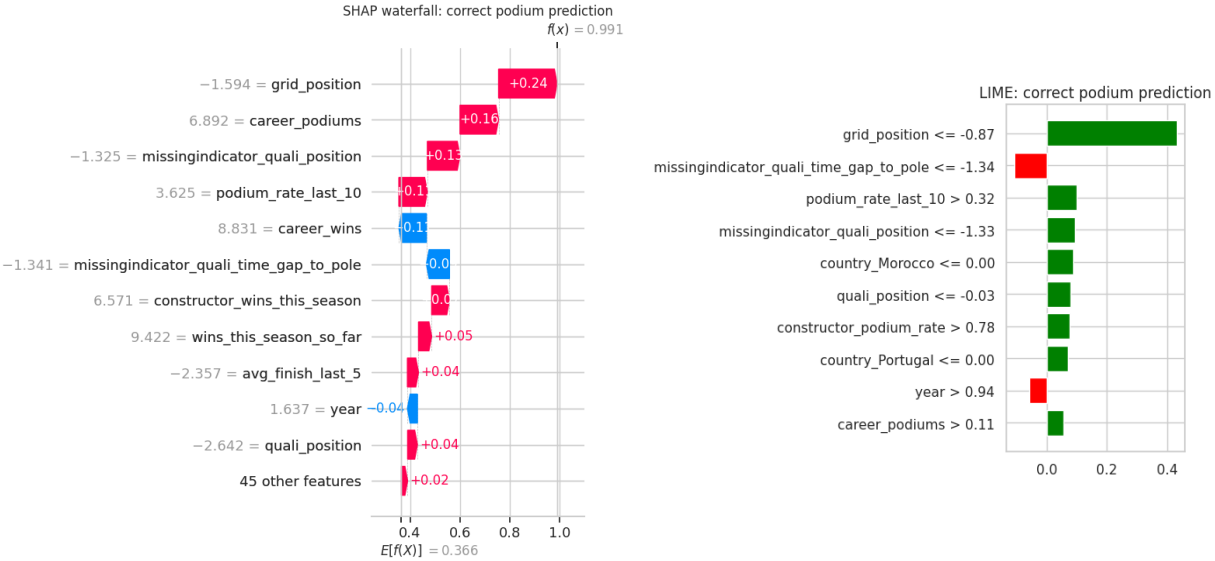


Figure 13. *Local explanation of the correct podium prediction: SHAP waterfall (left) and LIME (right).*

Missed podium: Carlos Sainz, 2021 Hungarian GP (started 15th, podium, predicted 0.077). Apart from the paired indicator columns (net about –0.08), grid 15 is the largest negative SHAP contribution (–0.11) and the largest LIME bar; his team's form pushes up only slightly. From pre-race information the prediction is reasonable. The race was reshaped by a first-lap pile-up, and Sainz was promoted to third only after a rival's post-race disqualification. Neither event can be known before the start, so the miss shows the limit of any pre-race model.

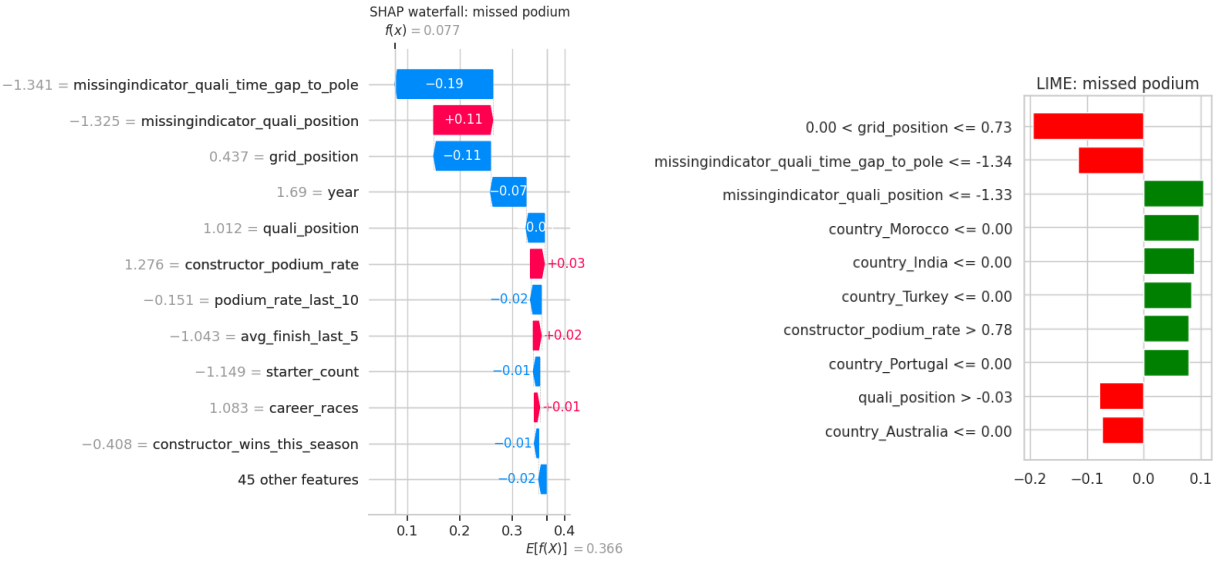


Figure 14. *Local explanation of the missed podium: SHAP waterfall (left) and LIME (right).*

LIME also shows rules such as "country_Morocco ≤ 0" (i.e. the race was *not* in Morocco). They reflect the model's real coefficients for rare countries (Morocco hosted a single race, in 1958) but are irrelevant to these records: LIME perturbs one-hot columns independently and creates unrealistic rows. SHAP gives them no weight because none of its background rows has those countries, which is why we read the two methods together.

8. Inference function

`predict_podium(race_data)` takes a dictionary of pre-race features and returns a podium probability from the selected model; missing fields are filled by the fitted imputer and flagged by the indicator columns. For a front-row starter from the test seasons, the complete record gives **0.953**. Without the qualifying position, the gap to pole and the recent average finish, it gives **0.907**: still above the 0.65 threshold, but a little less certain.

9. Known limitations

- **Qualifying-coverage drift:** qualifying data is complete only from 2003, so for most training rows the qualifying features are imputed and the model relies on the grid.
- **Era sensitivity:** cars, field sizes, points systems and reliability changed across 75 seasons. Training on 1950–2019 and testing on the 2022–2024 regulation era measures performance under drift, which is visible in the validation-to-test drop.
- **Small validation set:** two COVID-affected seasons (759 rows) make validation scores, and the model choice based on them, noisy. Hence the bootstrap-based selection rule.
- **Population choices:** pit-lane starters (grid 0) and excluded drivers are left out of the modelling table, and an exclusion can occasionally be decided after the race.
- **Team identity:** rebranded constructors get new IDs, which resets their form features.
- **Correlated features** make linear-model coefficients and explanations harder to read (Section 7).
- **Explanations are not causal.**

Appendix: additional figures from the notebook

The brief asks for all plots in both the notebook and the report. The figures below complete the set shown in the main text; their discussion is in the corresponding notebook sections.

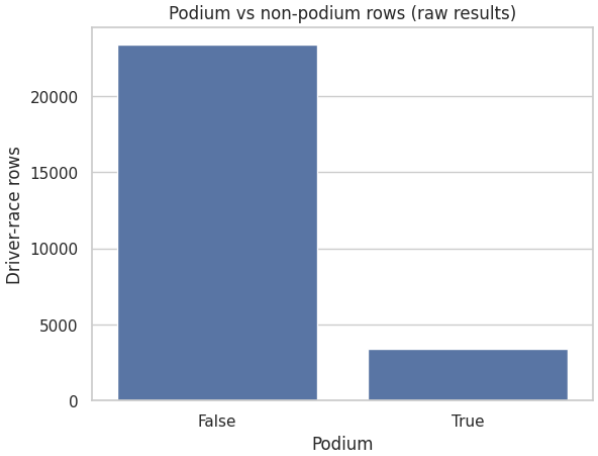


Figure 15. *Podium vs non-podium result rows in the raw data (12.7% podiums).*

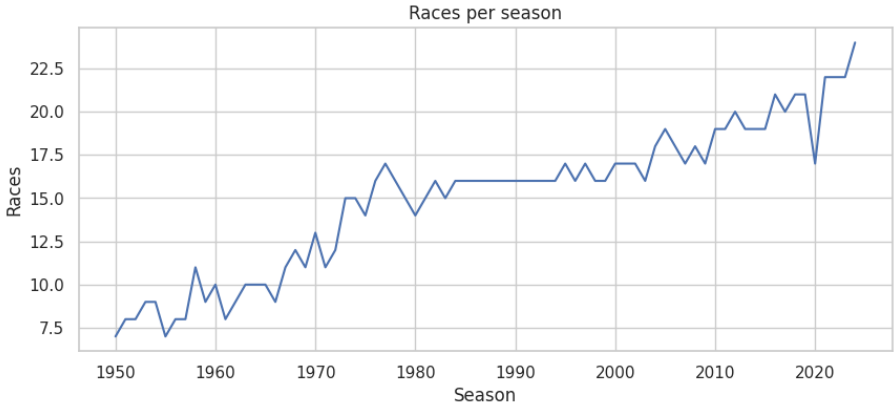


Figure 16. *Races per season.*

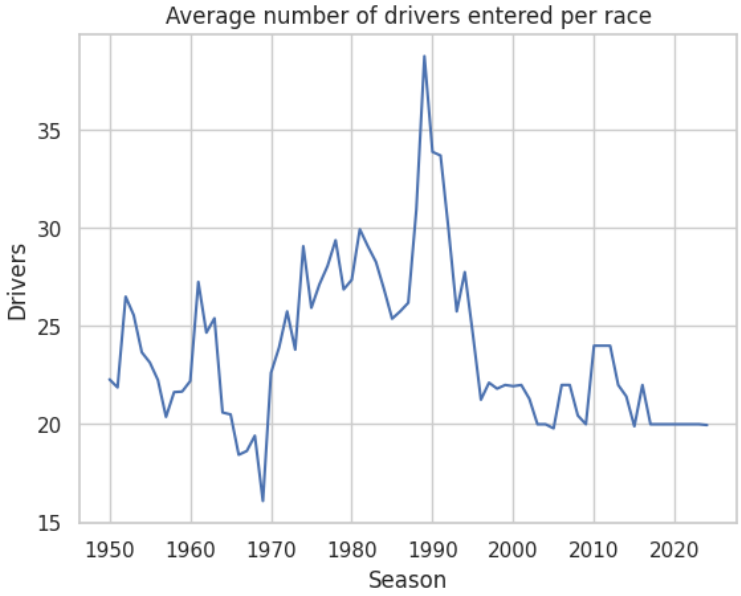


Figure 17. *Average number of drivers entered per race.*

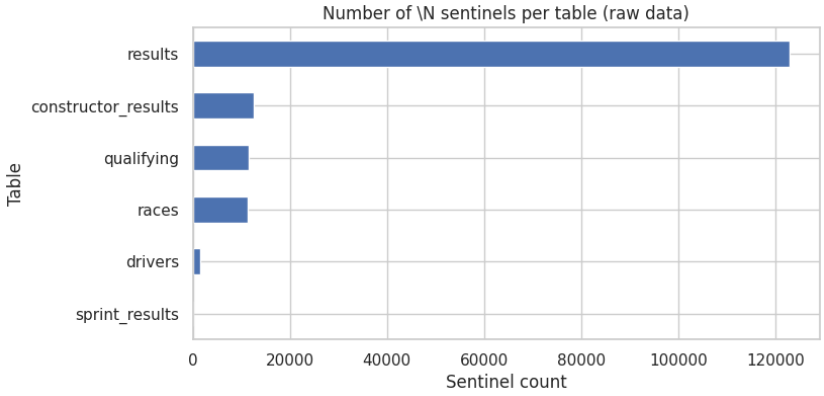


Figure 18. *Number of \N sentinels per table in the raw data.*

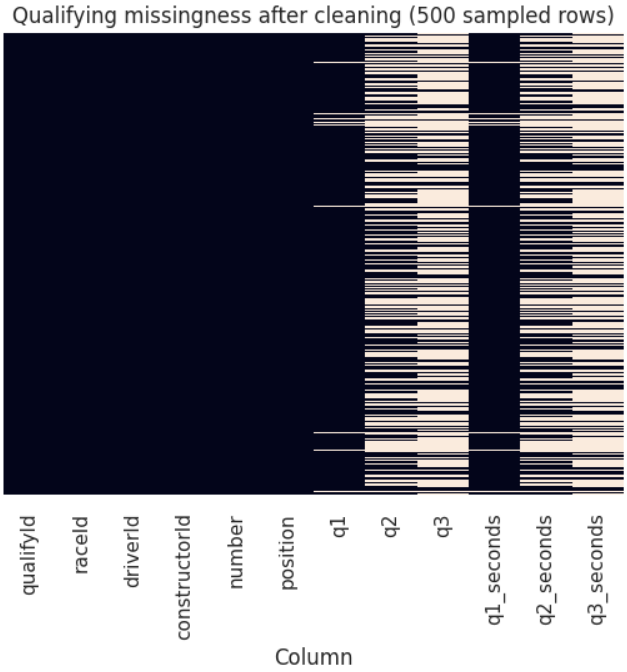


Figure 19. *Qualifying missingness after cleaning (500 sampled rows).*

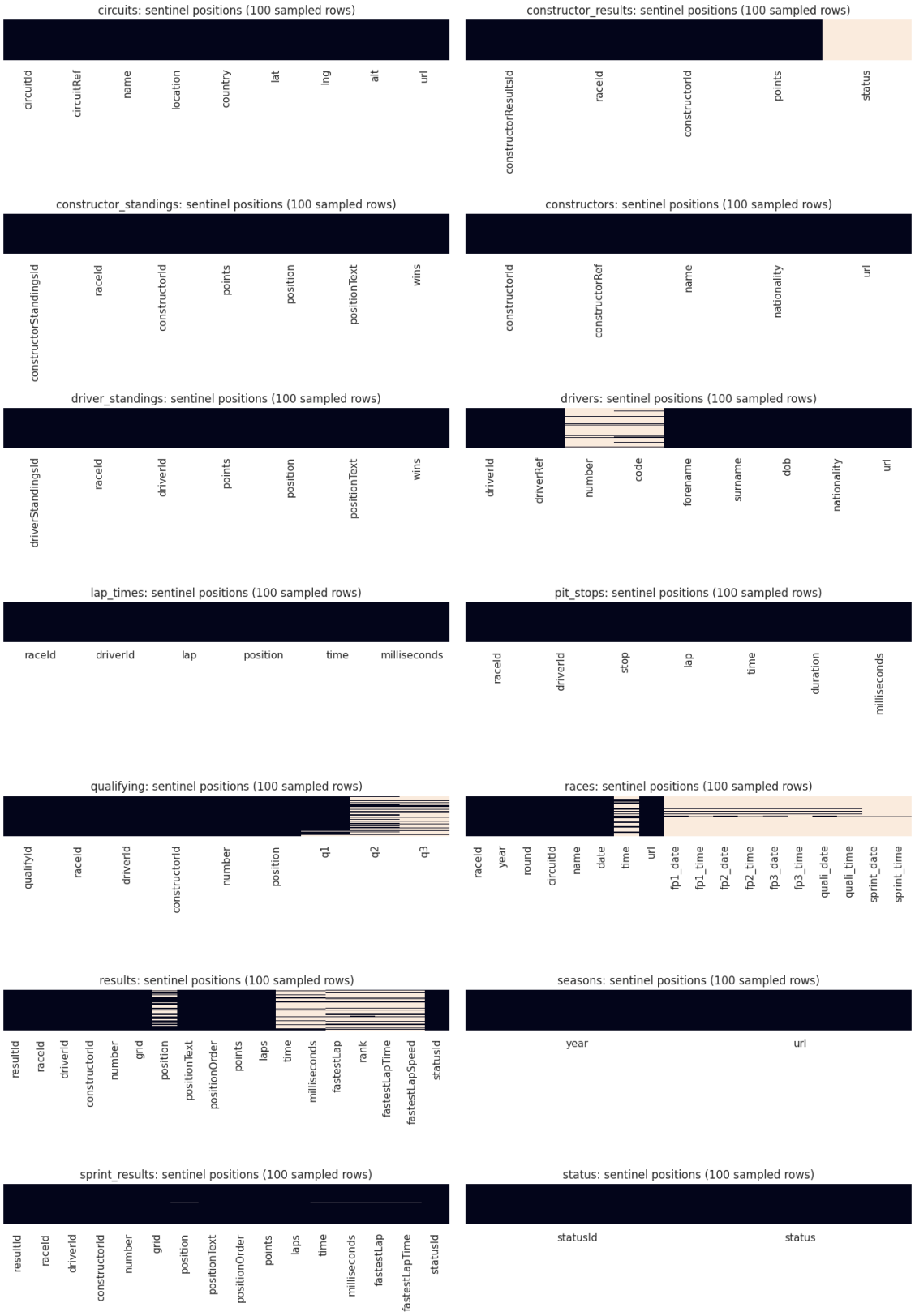


Figure 20. *Positions of \N sentinels in 100 sampled rows of every raw table.*

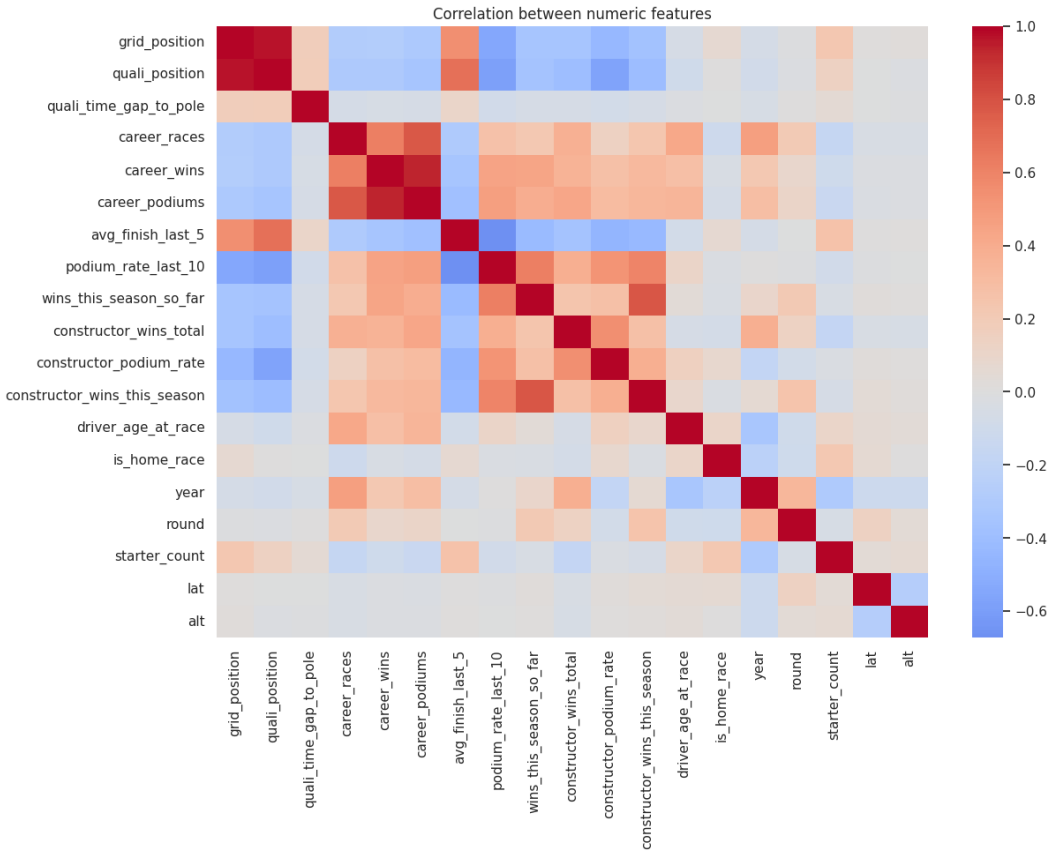


Figure 21. *Correlation between the numeric features (modelling table).*

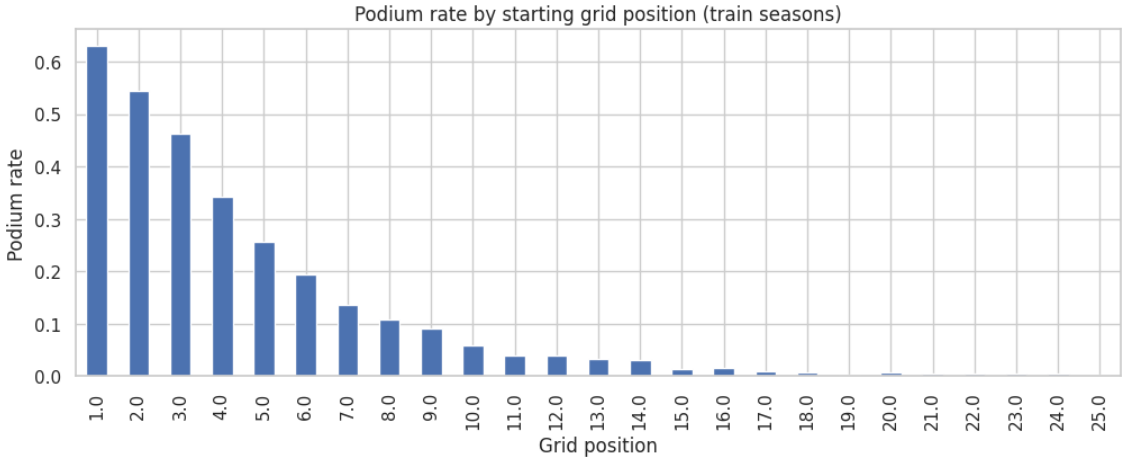


Figure 22. *Podium rate by starting grid position (train seasons).*

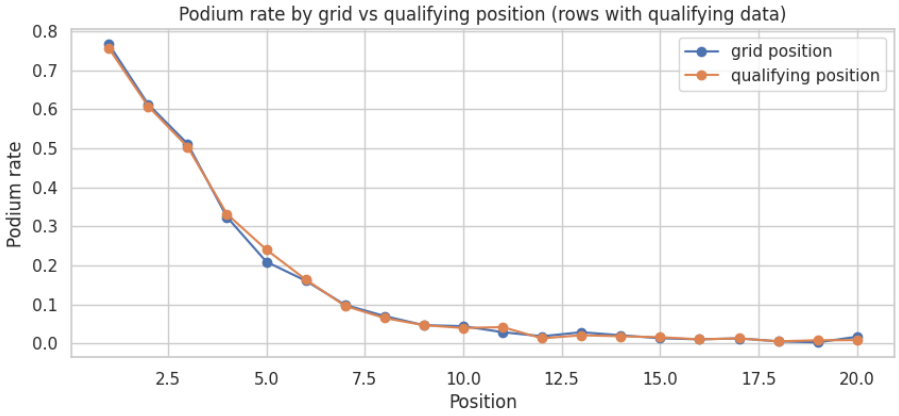


Figure 23. *Podium rate by grid vs qualifying position (rows with qualifying data).*

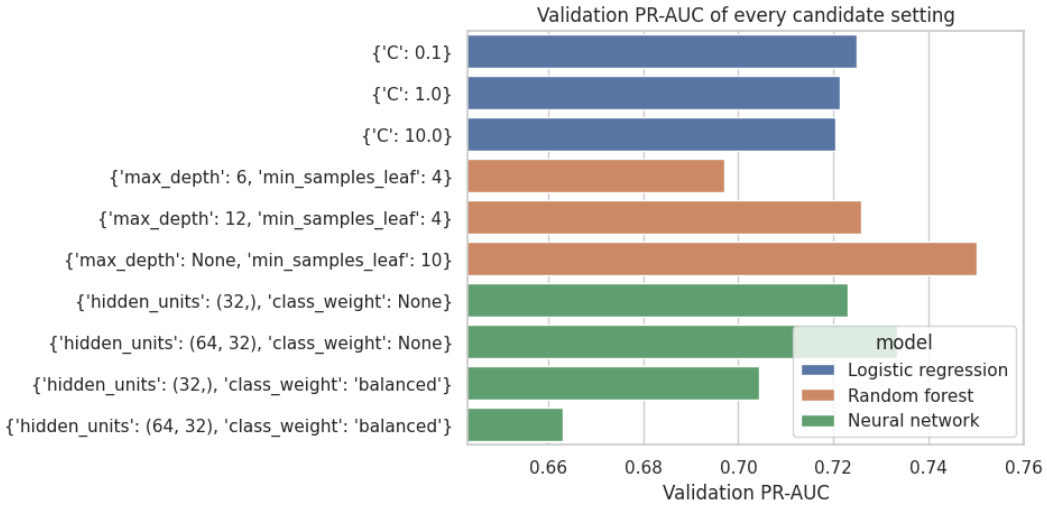


Figure 24. *Validation PR-AUC of every candidate hyperparameter setting.*

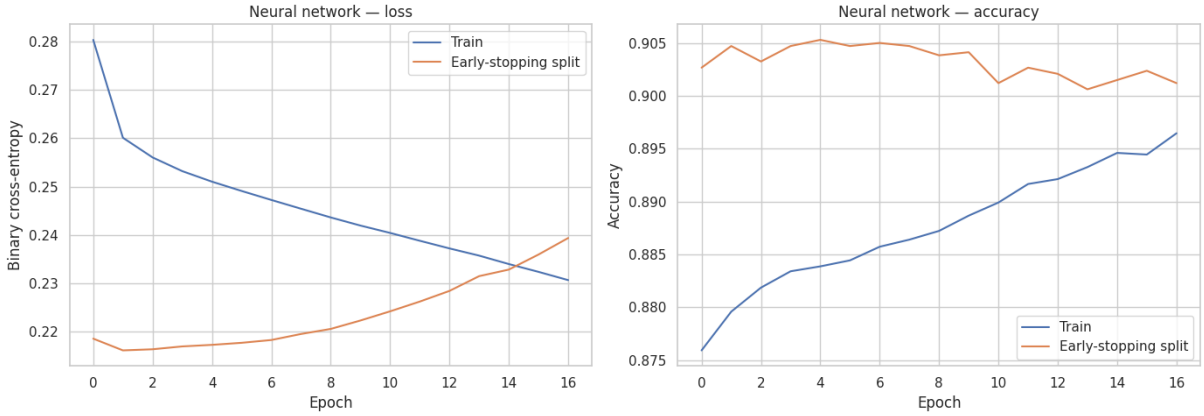


Figure 25. *Neural network training history; the early-stopping loss is lowest at the 2nd epoch and those weights were restored.*

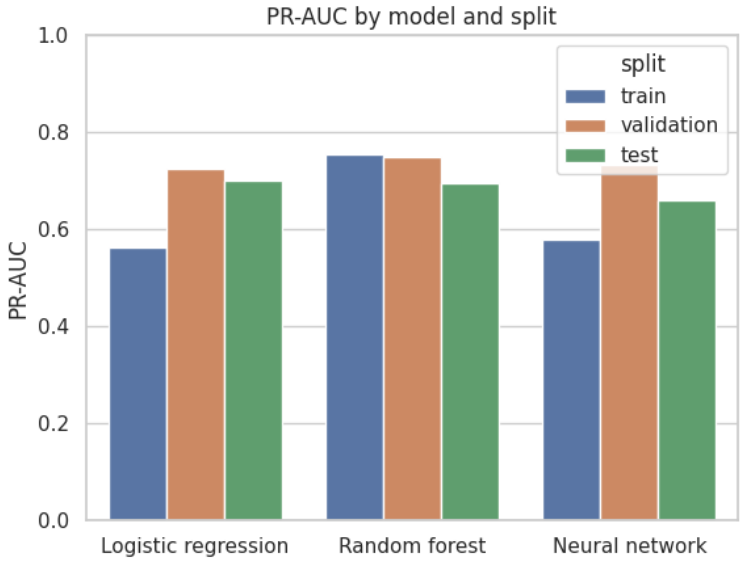


Figure 26. *PR-AUC of the three models on train, validation and test.*

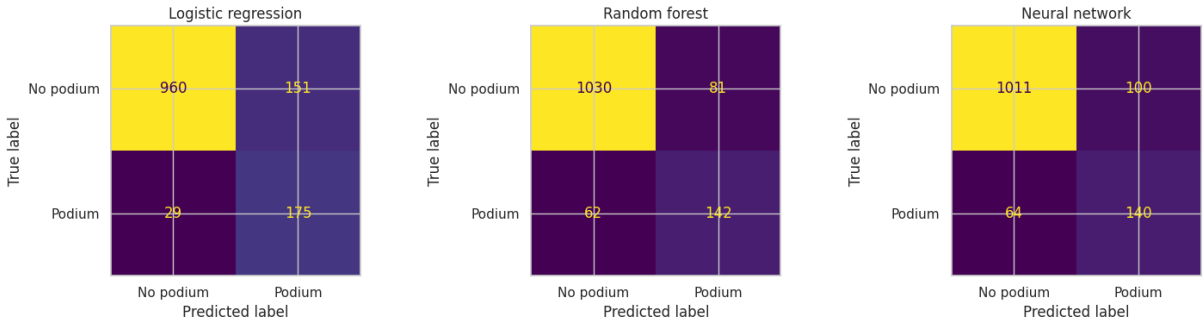


Figure 27. *Test-set confusion matrices at the validation-tuned thresholds.*